Java has 8 primitive data types; char, boolean, byte, short, int, long, float, and double. For this exercise, we'll work with the primitives used to hold integer values (byte, short, int, and long):

- A byte is an 8-bit signed integer.
- A short is a 16-bit signed integer.
- An int is a 32-bit signed integer.
- A long is a 64-bit signed integer.

Given an input integer, you must determine which primitive data types are capable of properly storing that input.

To get you started, a portion of the solution is provided for you in the editor.

Reference:

https://docs.oracle.com/javase/tutorial/java/nutsandbolts/datatypes.htm

Input Format

The first line contains an integer, T, denoting the number of test cases.

Each test case, T, is comprised of a single line with an integer, n, which can be arbitrarily large or small.

Output Format

For each input variable n and appropriate primitive $dataType$, you must determine if the given primitives are capable of storing it. If yes, then print:

```
n can be fitted in:
* dataType
```

If there is more than one appropriate data type, print each one on its own line and order them by size (i.e.: $byte < short < int < long$).

If the number cannot be stored in one of the four aforementioned primitives, print the line:

```
n can't be fitted anywhere.
```

Sample Input

```
5
-150
150000
1500000000
213333333333333333333333333333333333
-100000000000000
```

Sample Output

```
-150 can be fitted in:
* short
```

Change Theme Language Java 7

```
1  import java.util.*;
2  import java.io.*;
3
4
5
6  class Solution {
7      public static void main(String []argh)
8      {
9          Scanner sc = new Scanner(System.in);
10         int t=sc.nextInt();
11         for(int i = 0; i < t; i++)
12         {
13             try
14             {
15                 long x=sc.nextLong();
16                 System.out.println(x+" can be f
                                    Line: 35 Col: 1
```

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Test against custom input